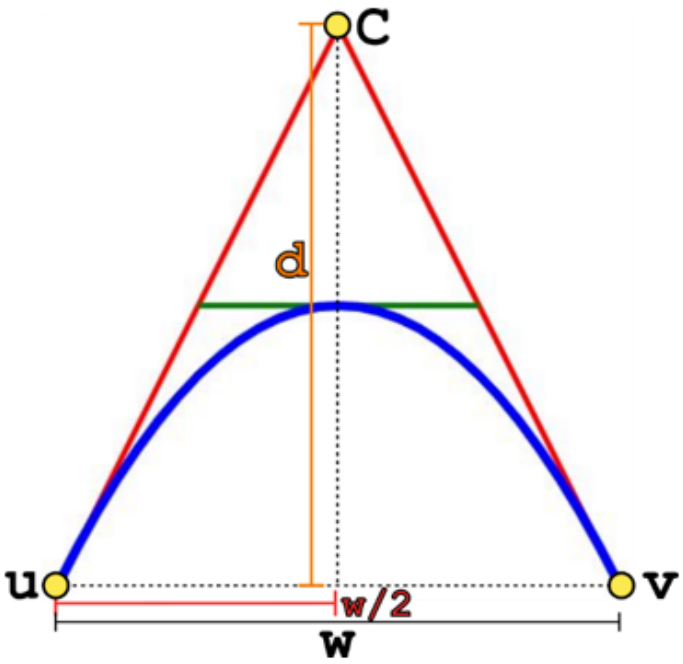


Figure 16.6 A quadratic Bézier curve used to draw an edge between vertices u and v. The point c is the curve's control point. For this example, we restrict to a subset of all the possible quadratic Bézier curves between two points, and in particular to just those curves where the control point lies on the line perpendicular to the segment uv, and passing through the middle of the segment (in other words, the line made by points in the plane that are at the same distance to u and v.

In particular, to stay true to the choice of a fully-randomized algorithm, we could just randomly choose each edge's control point(s); so much freedom in the choice, however, could lead to weird shapes for the edges, making it slower the convergence of the algorithm to a good solution.

A more feasible alternative could be restricting even more the curves that can be used, for example using quadratic curves whose control point is at the same distance from both edge's endpoints: this choice, explained in figure 16.6, allows us to